

Fig. 6: Screenshot of Keil firmware

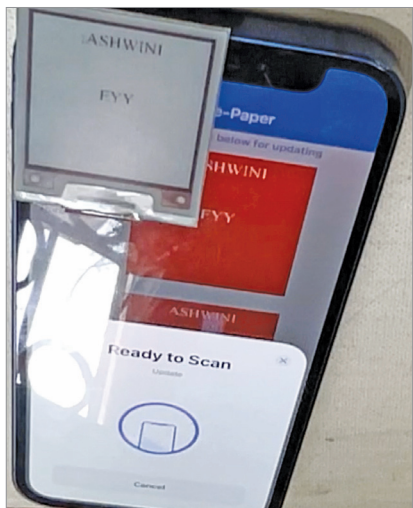


Fig. 7: Android phone showing the e-link display

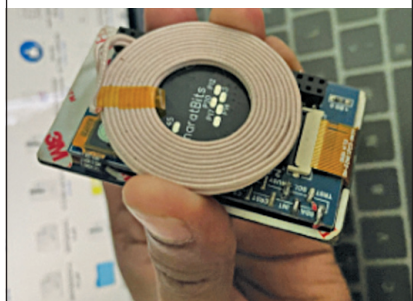


Fig. 8: Wireless energy harvesting coil with e-link display

TABLE 2
BILL OF MATERIALS FOR WI-FI BASED BATTERYLESS DISPLAY

Component	Description	Quantity
Indusboard coin (MOD1)	Coin sized board	1
Wireless energy harvesting coil (C1)	XKT-L10 20mm20uH coil	1
E-Ink display (MOD2)	2.13-inch SPI E-Ink display	1

denfc-m01-nfc-eink-display-board).

Before connecting the components, refer to Figs. 3, 4, and 5. First, connect the NFC energy harvesting GDEY0213B74 e-ink display driver module using the FPC connector provided on the DENFC-M0 energy harvesting board. Now, connect the ST emulator to the NFC display driver board. Fig. 3 shows the FPC connector and connection with the display. Internally, the display pin connections are shown in Fig. 4, while Fig. 5 displays the connection with the ST link emulator.

The firmware can be uploaded to the driver module using Keil. The board supports SWD download mode. Download the Keil and firmware, and then place them in the root catalogue of the Keil software. Select the MCU as ARMCM0. Now, compile the firmware and upload it to the board. Fig. 6 displays a screenshot of the Keil firmware.

For testing, download the app provided with the driver module's QR code and install it on your phone. Place the NFC display driver on the NFC side of your phone and upload the desired image or text. After a few seconds, the display will refresh and show the updated screen with the text and image you just uploaded. Fig. 7 illustrates an Android phone displaying the e-ink display.

Interestingly, there is no battery connected to the display driver, yet the entire display updates without any power source, making it batteryless. The NFC driver can be

removed from the phone, and the display will remain the same even without power. Additionally, the display can be detached from the driver module and placed wherever necessary, such as on an ID card, shop price list, or advertisement display; it will remain functional for years without needing any power.

Wi-Fi based batteryless display

To design a long-range wireless updatable and batteryless display, the Indusboard and a wireless powering coil can be used. As mentioned earlier, e-ink displays require only a small amount of power when updating, and the displayed screens remain unaltered even when power is removed. Therefore, the same concept as the earlier design is employed, but instead of NFC coil energy harvesting, energy is harvested from the magnetic flux using the coil. The Indusboard, wireless power transfer coil, a wireless charger module, and an e-ink display are utilised.

Whenever you want to update the display, simply place it on any wireless charger or charging module. The board receives power by harvesting the energy emitted from the charger's coil, and the display provides Wi-Fi connectivity. The display can then be connected to a smartphone or laptop to send the desired text or image to be shown on the screen. Once the screen is updated, it can be removed from the coil.

To embark on the Wi-Fi-based batteryless display, the components